

Lab: Predictive Coding in Neural Circuits

Learning Goal: Map the predictive coding architecture onto specific neural circuits and experimental data.

Part 1: Canonical Microcircuit Diagram

Exercise: Draw or describe the canonical microcircuit, labeling:

- Cortical layers 1-6
- Where predictions are generated (deep layers 5/6)
- Where prediction errors are computed (superficial layers 2/3)
- Where external input arrives (layer 4)
- Arrows showing top-down (prediction) and bottom-up (error) flow

Write your response here

Part 2: MMN Experiment Design

Learning Goal: Design a mismatch paradigm to test predictive coding predictions.

Exercise: Design a simple oddball experiment for visual processing (not auditory):

1. What repeated stimulus would you use (the "standard")?
2. What deviant stimulus would you insert (the "oddball")?
3. Where would you expect prediction error signals to originate in the brain?
4. What would you measure (EEG, fMRI)? What pattern would confirm predictive coding?

Write your response here

Part 3: Gain Control and Attention

Learning Goal: Understand precision as gain control through case analysis.

Scenarios: For each, describe what happens to neural gain in the relevant sensory areas:

1. You hear your name called in a noisy restaurant
2. A doctor carefully examines an X-ray, looking for a small tumor
3. You're driving and suddenly a truck brakes hard in front of you
4. You're daydreaming in a lecture and suddenly the professor says your name

Write your response here

Part 4: Visual Hierarchy Analysis

Learning Goal: Trace prediction and error flow through the visual system.

Exercise: You see a cat sitting on a windowsill for the first time vs. the hundredth time.

Visual Area	First Time: Prediction Error?	Hundredth Time: Prediction Error?	What changes?
V1 (edges)			
V2 (textures)			
V4 (shapes)			
IT (objects)			

Explain why neural activity decreases with familiarity (repetition suppression).

Write your response here

Part 5: Reflection

In 150 words, reflect: If the brain only passes prediction errors upward, does that mean consciousness is primarily about noticing what's unexpected? What are the implications for mindfulness or meditation?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Circuit diagramming	Canonical microcircuit architecture
2	Experiment design	Mismatch paradigms for visual system
3	Case analysis	Gain control and attentional precision
4	Hierarchical reasoning	Repetition suppression in visual hierarchy
5	Philosophical reflection	Consciousness and prediction error